Biology GCE O Level 2024 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0510 Biology 1, June 2024, Ordinary Level, General Certificate of Education Board). Duration: one and a half hours. Answer ALL 50 questions; all questions carry equal marks. Each question has four suggested answers A, B, C and D; select only one answer for each question. Non-programmable calculators are allowed.

Multiple choice questions - choose the one correct answer, A, B, C or D. Section I (questions 1-7): independent questions. Section II (questions 8-20): biological diagrams. Section III (questions 21-25): one or more responses correct. Section IV (questions 26-30): pairs of statements. Section V (questions 31-40): lettered headings and a description. Section VI (questions 41-50): biological situations.

1. *SECTION I - directions as printed, summarised:* Each of these questions or incomplete statements is followed by four suggested answers. Select the best answer in each case.

Which of the following organisms has all the following parts: vacuole, cell membrane, chloroplast and pyrenoids?

a) Bacteria
b) Amoeba
c) Yeast
d) Spirogyra

2. If a lake is polluted by untreated sewage, it would turn to have:

a) More algae.
b) More oxygen.
c) More fish.
d) More plants.

3. Which structure assists the amoeba both in food capture and movement?

a) Food vacuole.
b) Plasma membrane.
c) Pseudopodium.
d) Contractile vacuole.

4. Is a pain receptor on the skin of man:

a) Pacinian corpuscle.
b) Hair plexus.
c) Meissner's corpuscle.
d) Free nerve ending.

5. Where does implantation take place in the female reproductive system?

a) Cervix.
b) Uterus.
c) Oviduct.
d) Fallopian tube.

6. The left side of the heart is thicker than the right side because it pumps blood:

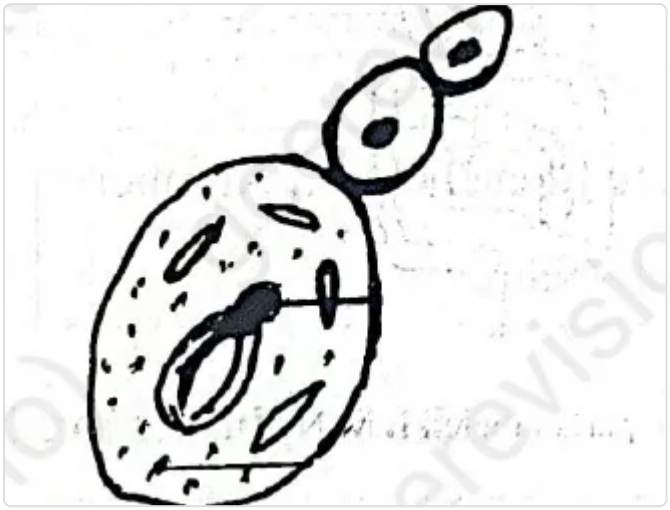
a) To shorter distances.
b) To the lungs only.
c) At a lower pressure.
d) To longer distances.

7. During gaseous exchange in humans, oxygen is sent into the lungs as a result of:

a) Increase in thoracic pressure and decrease in thoracic volume.
b) Decrease in thoracic volume and decrease in thoracic pressure.
c) Increase in thoracic volume and decrease in thoracic pressure.
d) Decrease in thoracic pressure and no change in volume.

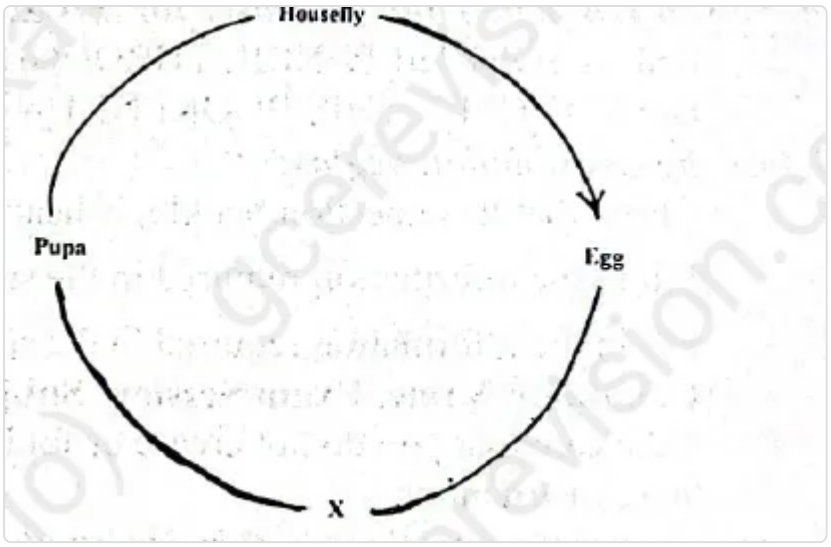
8. *SECTION II - directions as printed, summarised:* This group of questions relates to biological diagrams. Select the best answer for each question.

The mode of reproduction in the yeast above is



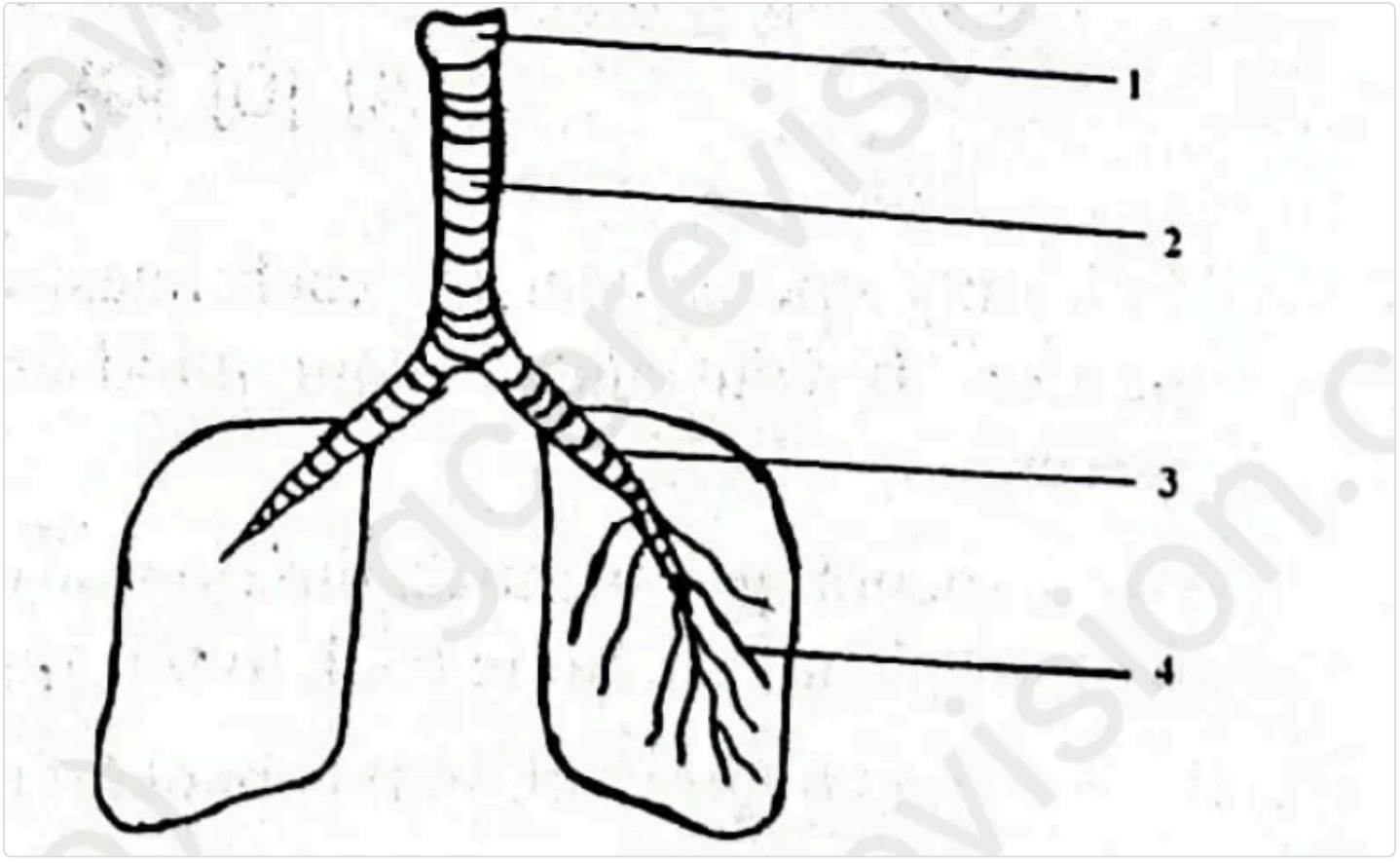
a) Conjugation.
b) Fragmentation.
c) Sporulation.
d) Budding.

9. The stage *"X"* on the life cycle below represents:



a) Imago.
b) Nymph.
c) Adult.
d) Larva.

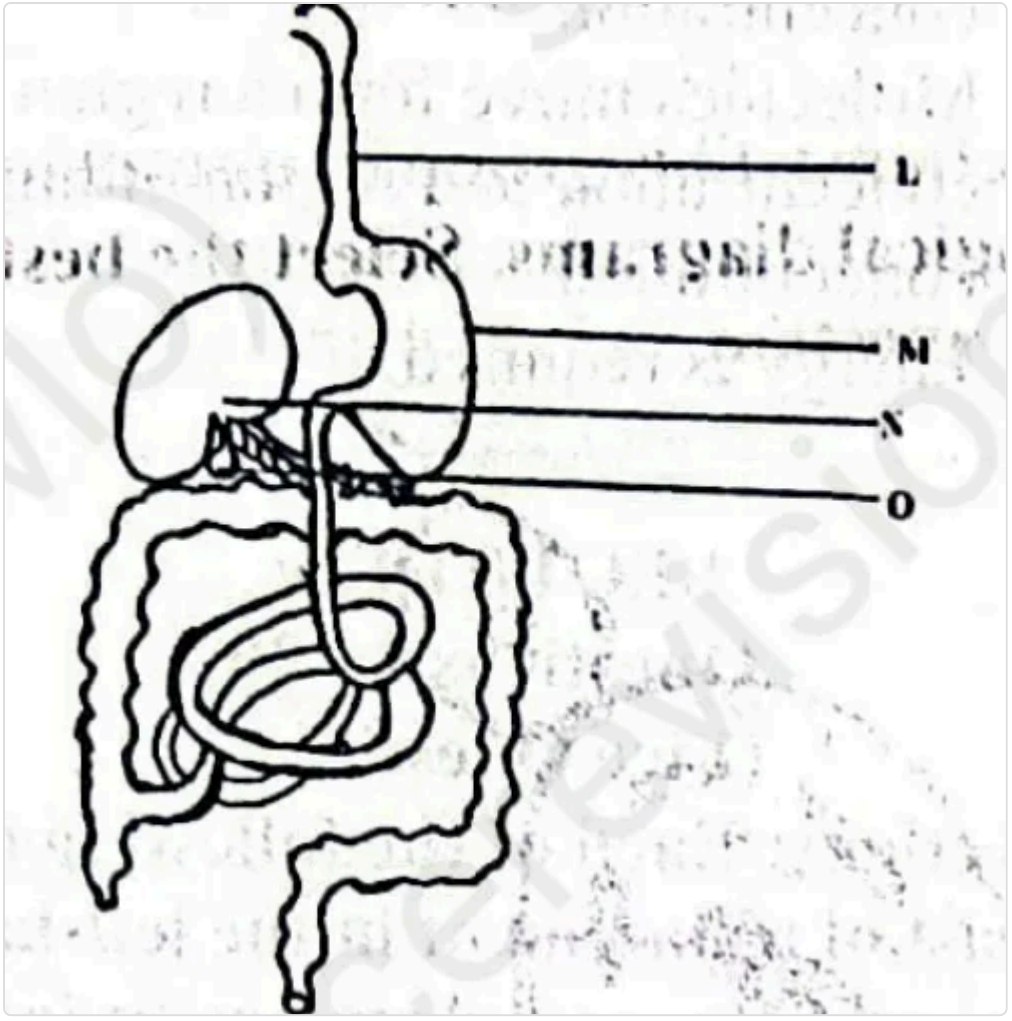
10. The diagram below is a cross section of part of the respiratory system in man.



The structure labelled 3 represents:

a) Bronchus.
b) Bronchioles.
c) Larynx.
d) Trachea.

11. The diagram below is that of the alimentary canal of man.



Identify the parts labeled L M N O.

a) L Oesophagus, M Stomach, N Liver, O Gall bladder
b) L Oesophagus, M Gall bladder, N Stomach, O Liver
c) L Stomach, M Liver, N Gall bladder, O Oesophagus
d) L Liver, M Gall bladder, N Oesophagus, O Stomach

12. The part of the leaf with numerous chloroplast is:

a) Palisade layer.
b) Spongy mesophyll.
c) Upper epidermis.
d) Stomata.

13. The part of the mammalian eye that refracts light first during image formation?

a) Pupil.
b) Vitreous humour.
c) Aqueous humour.
d) Cornea.

14. Where does meiosis occur in the male reproduction system?

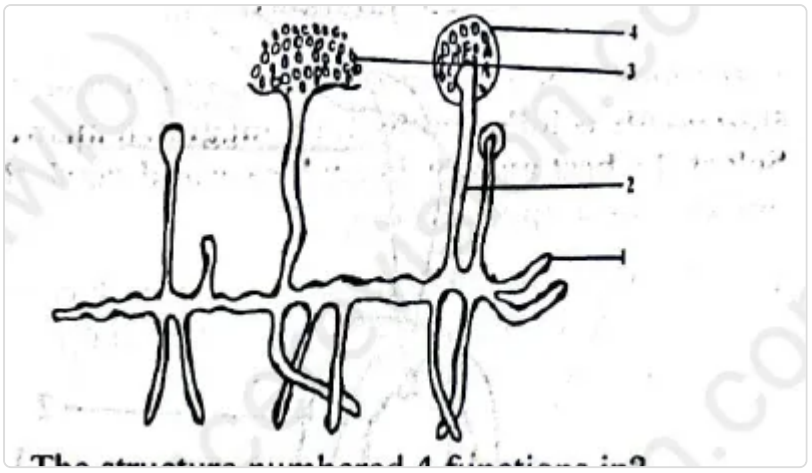
a) Penis.
b) Epididymis.

c) Scrotum.
d) Testes.

15. A structure which carries water from the roots to the leaves:

a) Cuticle.
b) Phloem.
c) Xylem.
d) Epidermis.

16. Below is the structure of a mould fungus:



The structure numbered 4 functions in:

a) Absorption of food.
b) Enclosure of spores.
c) Secretion of enzymes.
d) Sexual reproduction.

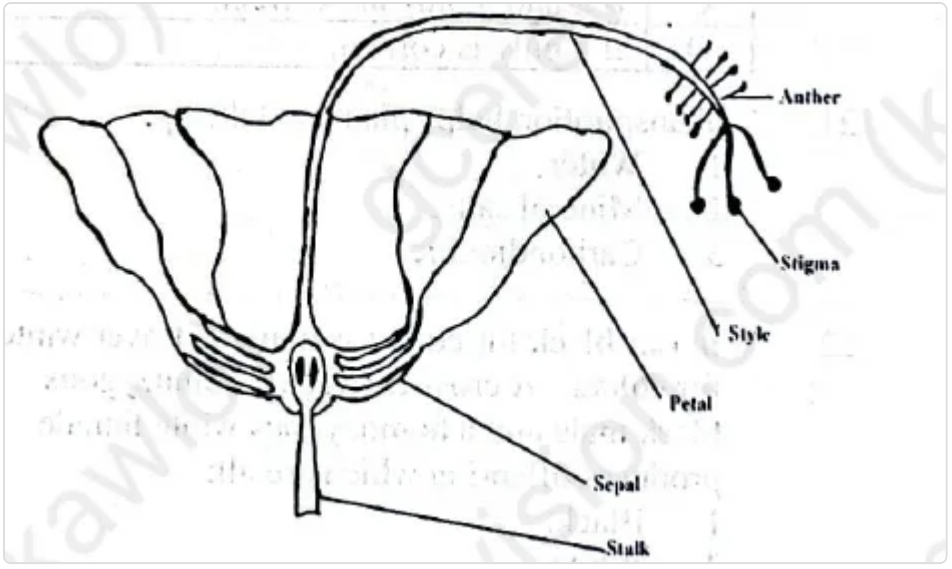
17. Which part of the mammalian brain is responsible for voluntary activities of the body?

a) Hypothalamus.
b) Cerebrum.
c) Cerebellum.
d) Medulla oblongata.

18. The sensory neuron plays a role in conducting impulses from the:

a) Receptor to the central nervous system.
b) Central nervous system to the sensory organ.
c) Effector to the central nervous system.
d) Sensory receptor to the motor neuron.

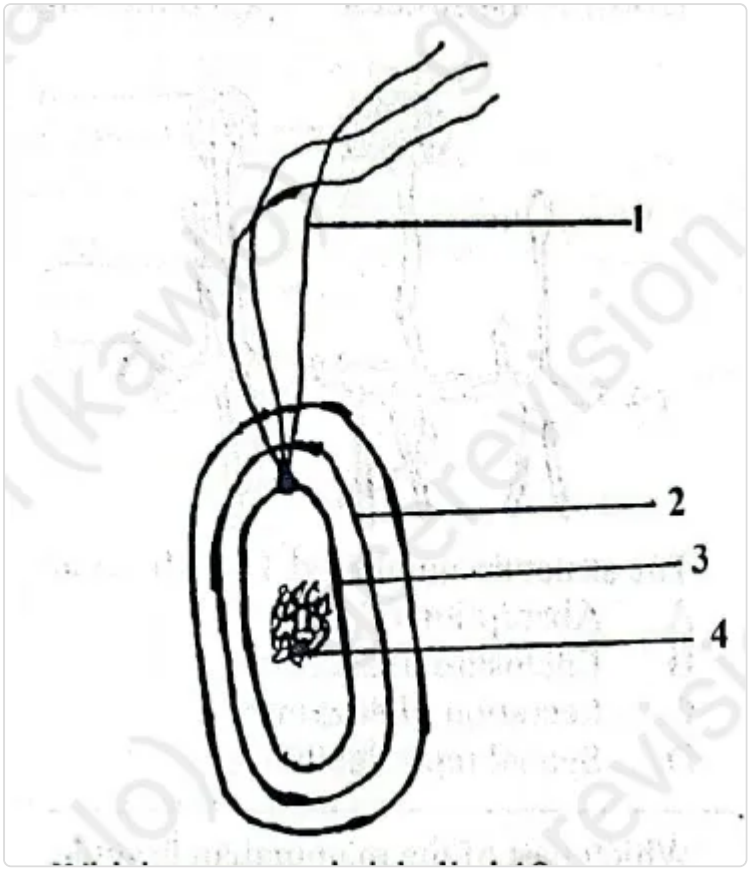
19. The diagram below is a cross-section of a hibiscus flower



Which part is responsible to attract insects for pollination?

a) Petal.
b) Stigma.
c) Anther.
d) Sepal.

20. Below is a diagram of a bacterial cell.



Which structure is labelled 1?

a) Flagellum.
b) Mucus layer.
c) Cell wall.
d) Cell membrane.

21. *SECTION III - directions as printed, summarised:* For each of the questions below, ONE or MORE of the responses is (are) correct. Decide which of the responses is (are) correct. Then choose: A if 1, 2 and 3 are all correct; B if 1 and 2 only are correct; C if 2 and 3 only are correct; D if 1 only is correct.

Transpiration helps plants to take up:

1. Water.
2. Mineral salts.
3. Carbondioxide.

a) If 1, 2 and 3 are all correct.
b) If 1 and 2 only are correct.
c) If 2 and 3 only are correct.
d) If 1 only is correct.

22. In rats black fur colour is dominant over white fur colour. A cross between a homozygous black male and a homozygous white female produces offspring which are all:

1. Black.
2. White.
3. Black and white.

a) If 1, 2 and 3 are all correct.
b) If 1 and 2 only are correct.
c) If 2 and 3 only are correct.
d) If 1 only is correct.

23. Reptiles:

1. Are poikilothermic.
2. Have moist scaly skin.
3. Are all terrestrial.

a) If 1, 2 and 3 are all correct.
b) If 1 and 2 only are correct.
c) If 2 and 3 only are correct.
d) If 1 only is correct.

24. When there is overheating in the body:

1. More blood flows to the skin surface.
2. Less blood flows to the skin surface.
3. There is an increase in metabolic rate.

a) If 1, 2 and 3 are all correct.
b) If 1 and 2 only are correct.
c) If 2 and 3 only are correct.
d) If 1 only is correct.

25. During active transport:

1. Molecules move from a region of high concentration to a region of low concentration.
2. Molecules move form a region of lower concentration to a region of high concentration.

3. Energy is required.

a) If 1, 2 and 3 are all correct.
b) If 1 and 2 only are correct.
c) If 2 and 3 only are correct.
d) If 1 only is correct.

26. *SECTION IV - directions as printed, summarised:* Each question consists of a statement in the left-hand column followed by a second statement in the right-hand column. Choose: A if both are true and the 2nd statement is a correct explanation of the first; B if both are true and the 2nd statement is NOT a correct explanation of the first; C if the first is true and the second false; D if the first is false and the second true.

First statement: Photosynthesis occurs only during the day.

Second statement: Chlorophyll absorbs solar energy in sun light.

Options as printed: A) True, True: the 2nd statement is a correct explanation of the first. B) True, True: the 2nd statement is NOT a correct explanation of the first. C) True, False. D) False, True.

First statement: Yeast cells carry out fermentation.

Second statement: Ethanol is a product of anaerobic respiration in breweries.

Options as printed: A) True, True: the 2nd statement is a correct explanation of the first. B) True, True: the 2nd statement is NOT a correct explanation of the first. C) True, False. D) False, True.

First statement: Nitrogen fixing bacteria converts soil nutrients to atmospheric nitrogen.

Second statement: Nitrogen fixation is either done aerobically or anaerobically.

a) True, True: the 2nd statement is a correct explanation of the first.
b) True, True: the 2nd statement is NOT a correct explanation of the first.
c) True, False.
d) False, True.

First statement: During very cold weather, the erector pili muscles contract and cause hair on the body to stand erect.

Second statement: During cold weather there is increase in metabolic activities which causes more blood supply to the skin surface.

a) True, True: the 2nd statement is a correct explanation of the first.
b) True, True: the 2nd statement is NOT a correct explanation of the first.
c) True, False.
d) False, True.

First statement: Plants are autotrophic.

Second statement: Plants always occupy the first trophic level in an ecosystem.

Options as printed: A) True, True: the 2nd statement is a correct explanation of the first. B) True, True: the 2nd statement is NOT a correct explanation of the first. C) True, False. D) False, True.

31. *SECTION V - directions as printed, summarised:* Each question consists of four lettered headings A, B, C, D. Select the heading which is closely related to the description which follows.

Select the heading which is closely related to the description: an example of a joint that can rotate 360°.

a) Hinge joint.
b) Ball and socket joint.
c) Fixed joint.
d) Slightly moveable joint.

32. Select the heading which is closely related to the description: a feeding relationship between two organisms in which one benefits but the other neither benefits nor loses.

a) Commensalism.
b) Parasitism.
c) Mutualism.
d) Symbiosis.

33. Select the heading which is closely related to the description: provides turgidity to plant cells.

a) Cell membrane.
b) Cell wall.
c) Chloroplast.
d) Nucleus.

34. Select the heading which is closely related to the description: a water pollutant.

a) Carbondioxide.
b) Smog.
c) Fertilizer.
d) Sulphurdioxide.

35. Select the heading which is closely related to the description: a primary consumer.

a) Herbivore.
b) Omnivore.
c) Carnivore.
d) Parasite.

36. Select the heading which is closely related to the description: a condition in which alleles blend to produce an intermediate trait.

a) Phenotype.
b) Genotype.
c) Complete dominance.
d) Incomplete dominance.

37. Select the heading which is closely related to the description: separates the right auricle from the right ventricle.

a) Bicuspid valve.
b) Tricuspid valve.
c) Semi lunar valve.
d) Septum.

38. Select the heading which is closely related to the description: a product of fermentation in plants.

Options as printed: A) Carbondioxide. B) Water. C) Ethanol. D) Lactic acid.

39. Select the heading which is closely related to the description: detects vibrations in water.

a) Scales.
b) Lateral line.
c) Gills.
d) Tail fin.

40. Select the heading which is closely related to the description: an adaptation of birds to flight.

a) Low body temperature.
b) Sharp beak.
c) Streamline nature.
d) Small flat muscles.

41. *SECTION VI - directions as printed, summarised:* This group of questions deals with biological situations. Each situation is followed by a set of questions. Select the best answer for each question.

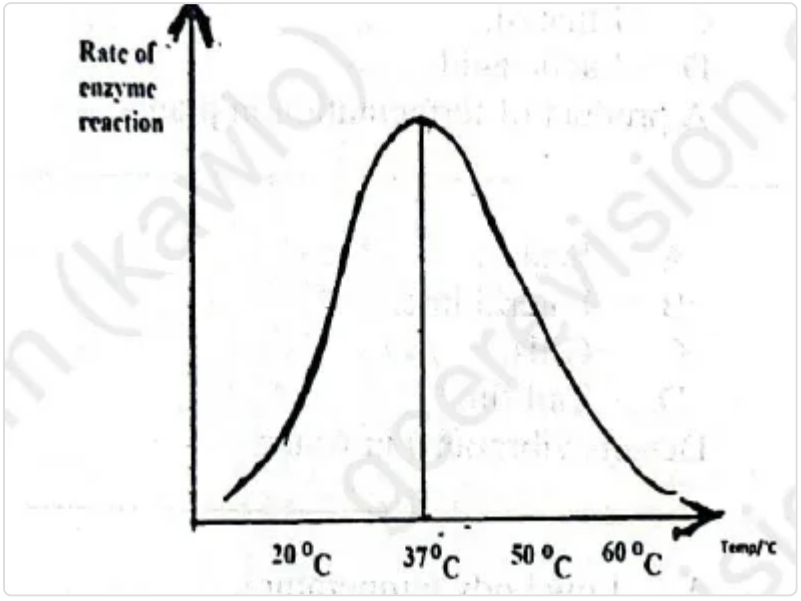
The bending of a plant shoot towards unidirectional light source is the result of:

a) Accumulation of auxins on the lighted side
b) A decrease in the amount of water in the cells of plant.
c) Accumulations of auxins on the darker side.
d) An increase in the rate of water loss from the leaves.

42. If a cell has 16 chromosomes, how many chromosomes will each of its daughter cells have after mitosis.

a) 4.
b) 8.
c) 12.
d) 16.

43. The graph below show the effect of temperature on an enzyme catalyzed reaction.



What will cause a fall in the curve at temperatures above 40°C.

a) Enzymes are deactivated.
b) Enzymes are denatured.
c) Enzymes are used up.
d) Enzymes are activated.

44. What is the use of pyrogallic acid in the experiment to demonstrate the importance of oxygen during seed germination?

a) To absorb all oxygen present.
b) To produce more oxygen.
c) To cause the seeds to germinate.
d) To provide more water for the seeds.

45. A seedling transplanted on a hot windy day withers because the:

a) Sun is too hot.
b) Air is too dry.
c) Seedling loses more water.
d) Roots were destroyed.

46. A solution from crushed germinating maize seeds was heated with an equal amount of Benedict's solution in a test tube and a brick red colour appeared. The brick red colour indicate the presence of:

a) Starch.
b) Sugar.
c) Lipid.
d) Protein.

47. A food chain differs from a food web in that:

a) Each organism in a food web has more than one food source.
b) Little energy is lost in a food web
c) Each organism in a food chain has more than one food source.
d) Food chains are interconnected.

48. In an experiment to test for the presence of starch in a green leaf, the leaf is boiled in water, then in alcohol. Why must the leaf be boiled in water?

Options as printed: A) To burst starch grains. B) To remove chlorophyll. C) To destroy the protoplasm. D) To make it soft.

49. In a farm land, it is noticed that maize plants found amongst leguminous plants, were seen to grow more vigorously than in areas where the leguminous plants were absent. How can this be explained?

a) The leguminous plants make more water available to the maize plants.
b) The leguminous plants provide shade to the maize plants.
c) The leguminous plants increase the nitrogen content of the soil.
d) The leguminous plants conserve more soil water.

50. The planting of more trees in an environment will cause:

a) Production of more CO2
b) Soil erosion.
c) Production of more oxygen.
d) Soil leaching.

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